

RESEARCH PROGRAM 2026

# Proofs, Reproducibility, and Open Questions

*AI self-reference · statistical design · information-theoretic physics · spectral rank · Navier-Stokes Fourier audits*

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*This edition includes work completed before the outcome of the ongoing N14 prospective run.*

# How to read this document

This PDF is a public-facing synthesis of a research lineage, not a replacement for the frozen source releases. When wording or numbers differ, the versioned primary artifact, manifest, hash record, or repository commit controls. The purpose here is to make the logic readable in one place while preserving the boundary between theorem, computation, simulation, falsification, and open questions.

**CLAIM LEVELS** PROVED = analytic result under stated assumptions. NUMERICAL CERTIFICATE = exhaustive or high-precision computation, not interval/formal proof. SIMULATION VALIDATION = operating-characteristic evidence under a specified data-generating process. PREREGISTERED = hypotheses and rules fixed before the relevant data. FALSIFIED = a proposed stronger claim was explicitly defeated by counterexample or audit. OPEN = not established.

## Executive map

| Research line                      | Strongest completed result                                                                                                               | Status boundary                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Self-Referential Processing v8.6.3 | Locked 260-item, two-domain, five-gate prospective protocol; 1,000-seed recovery harness passed                                          | Protocol/implementation validation; not a live consciousness result            |
| DesignLab Phase 7                  | 4,000 GLMM-fit simulation; M1 passed locked null calibration/coverage screens, M2 did not                                                | Simulation-only architecture selection for a future preregistration            |
| Physics v1.0 Gate 0                | Observer-entropy to horizon-entropy bridge remains unresolved under original assumptions                                                 | Negative audit / stop condition, not impossibility proof                       |
| Appendix A v1.4/v1.4.1             | Exact positivity theorem $l_1 > 0$ on the positive e-driven Hopf family                                                                  | Analytic theorem inside a phenomenological ODE; not a microscopic law          |
| LRSC odd-ring theorem v1.2         | $\text{rank}(V)=p$ for the defined single-step odd-ring operator under mask assumptions                                                  | Analytic finite-operator theorem; not a fluid or universal compression theorem |
| LRSC K0.001 supplement             | Fixed $M=99$ benchmark has no passing subset $K \leq 10$ ; a $K=11$ witness passes                                                       | Exhaustive IEEE-double numerical certificate for one benchmark                 |
| Navier-Stokes bridge / WP16        | Small-data H1 estimate; falsification of overstrong flux bounds; K36 N12/N13 prospective static holdouts plus post-hoc dynamical failure | No arbitrary-data 3D regularity proof; N14 result excluded from this edition   |

## Contents

- 1. Methodological spine: lock first, test second
- 2. Self-Referential Processing benchmark v8.6.3
- 3. DesignLab Phase 7 inferential validation
- 4. Physics v1.0 Gate 0: the entropy bridge audit
- 5. Appendix A: exact Hopf-bifurcation theorem
- 6. LRSC odd-ring spectral rank theorem
- 7. LRSC fixed-benchmark compression certificate
- 8. Navier-Stokes bridge audit: what survives and what fails
- 9. WP16 K36 finite mechanism and the N14 prospective gate
- 10. Public reproducibility ledger and references

# 1. Methodological spine: lock first, test second

Across otherwise different subjects, the research program converged on one discipline: define the claim before looking at the evidence that will judge it. The same discipline appears in the AI preregistration, the DesignLab simulation, the LRSC exhaustive audits, and the Navier-Stokes N12/N13/N14 holdouts.

- Freeze definitions, estimands, thresholds, seeds, and scope before confirmatory evaluation.
- Separate analytic proof from numerical corroboration; neither substitutes for the other.
- Keep failed or inconvenient results in the record rather than retuning after seeing outcomes.
- Use counterexamples aggressively to kill claims that are stronger than the mathematics supports.
- Record hashes, environment assumptions, and recovery incidents so a polished final artifact does not erase the path by which it was obtained.

**CORE PRINCIPLE** A reproducible negative result can be more scientifically valuable than a positive result obtained by moving the target after observation.

## 2. Self-Referential Processing benchmark v8.6.3

The Self-Referential Processing project operationalizes a narrow behavioral question: whether a model behaves differently when material is related to its own prior outputs under controls designed to remove stance, style, memorization, and scale explanations. The protocol explicitly does not define a test of subjective consciousness.

### 2.1 Frozen design

| Element          | Locked specification                                                                                                                                                              |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item pool        | 260 prompts; 52 in each of five categories; seven grammatical stems; no pre-written options                                                                                       |
| Domain 1         | Stance self-consistency: own harvested output versus a controlled perturbation, compared against peer-derived pairs                                                               |
| Domain 2         | Output authorship attribution under four controls; Control 4 (style-equalized + paraphrased + shuffled) is primary                                                                |
| Portfolio        | 20 models: 12 open-weights primary cells and 8 closed-API descriptive sensitivity cells                                                                                           |
| Seeds            | Position, tie-break, bootstrap, and peer assignment fixed at 20260910                                                                                                             |
| Gate 5           | Item-level linear probability model with two-way cluster-robust covariance on <code>model_id</code> and <code>item_id</code> ; <code>IsSelf</code> coefficient <code>beta3</code> |
| Gate 5 threshold | <code>beta3</code> $\geq 0.10$ and lower 95% confidence bound $> 0$ , fit on all 12 open-weights primary cells                                                                    |
| Survival floor   | At least 10/20 cells pass Gates 1-4, including at least six open-weights models                                                                                                   |

### 2.2 The five-gate falsification architecture

1. Gate 1 - informational symmetry: a 90% TOST interval inside  $\pm 0.05$  nats/token.
2. Gate 2 - stylometric control: a style signal of at least 0.10 with lower 95% bound above zero after 50  $\pm$  5 word scrubbing.
3. Gate 3 - cross-domain SESOI: both behavioral domains require effects of at least 0.10 with lower 95% bounds above zero.

4. Gate 4 - stateless baseline: empirical advantage must beat a BM25 + bge-large-en-v1.5 retriever baseline by at least 0.05 with a clustered-bootstrap lower bound above zero.
5. Gate 5 - scale regression: the portfolio-wide IsSelf coefficient must meet the registered effect-size and confidence-bound threshold after controlling for context length, order, stance distance, and centered scale.

## 2.3 Recovery harness: implementation evidence, not scientific data

The v8.6.3 recovery harness generated synthetic long-format data using the same estimator path as the confirmatory analysis. Its purpose was to detect implementation failures before live collection. It was explicitly excluded from confirmatory effect estimates.

| Scenario      | Seeds | Bias    | Two-sided rejection | Directional check         |
|---------------|-------|---------|---------------------|---------------------------|
| beta3 = +0.15 | 1000  | +0.0097 | 100.0%              | 0.0% negative estimates   |
| beta3 = 0.00  | 1000  | +0.0105 | 6.3%                | 50.7% negative estimates  |
| beta3 = -0.15 | 1000  | +0.0097 | 100.0%              | 100.0% negative estimates |

**RESULT BOUNDARY** The recovery harness passed the registered implementation checks. That establishes that the estimator can recover synthetic signals under the tested generator. It does not establish a Self effect in real model behavior.

## 3. DesignLab Phase 7 inferential validation

DesignLab is a separate simulation line used to stress candidate analysis architectures before future prospective use. It does not retroactively change the frozen v8.6.3 Gate 5 model.

### 3.1 Locked simulation

- 500 deterministic replicates.
- Four data-generating cells crossing true beta\_Self in {0.00, 0.10 log-odds} with true model intercept-slope correlation rho in {0.00, 0.30}.
- Each generated cell contains 100 models x 400 items = 40,000 observations with exact within-model Self balance.
- Each dataset is fit twice: M1 uses correlated model intercept/Self-slope random effects; M2 uses the corresponding uncorrelated block. Both include an item random intercept.
- Total: 500 x 4 x 2 = 4,000 GLMM fits. Every M1/M2 pair is fit to the same generated dataset, verified by dataset hash.

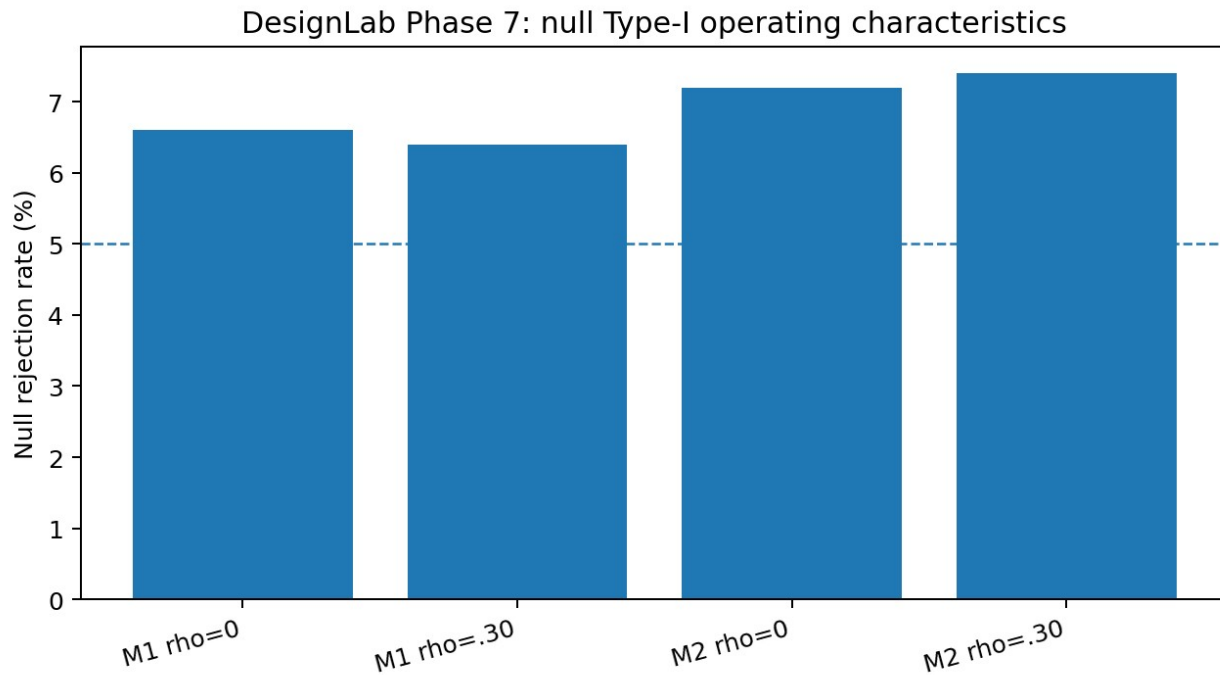


Figure 1. Phase 7 null rejection rates. The locked decision used exact-binomial compatibility tests, not visual proximity to 5%.

| Cell         | Spec | Reject | Coverage | Mean estimate | Exact null p | Decision |
|--------------|------|--------|----------|---------------|--------------|----------|
| Null rho=0   | M1   | 6.6%   | 93.4%    | 0.00027       | 0.1007       | PASS     |
| Null rho=0   | M2   | 7.2%   | 92.8%    | 0.00035       | 0.0305       | FAIL     |
| Null rho=.30 | M1   | 6.4%   | 93.6%    | 0.00019       | 0.1504       | PASS     |
| Null rho=.30 | M2   | 7.4%   | 92.6%    | 0.00028       | 0.0179       | FAIL     |

Under the prespecified rule, M1 passed both null Type-I and coverage compatibility screens, while M2 failed both. M1 therefore became the candidate architecture for a future preregistered design. The result is conditional on the simulated data-generating process and Monte Carlo resolution; nonrejection is not equivalence.

### 3.2 Effect recovery

| Condition          | Spec | Mean estimate | Bias     | Rejection | Coverage |
|--------------------|------|---------------|----------|-----------|----------|
| beta=0.10, rho=0   | M1   | 0.100405      | 0.000405 | 99.6%     | 94.2%    |
| beta=0.10, rho=0   | M2   | 0.100485      | 0.000485 | 99.8%     | 93.4%    |
| beta=0.10, rho=.30 | M1   | 0.100415      | 0.000415 | 99.6%     | 94.2%    |
| beta=0.10, rho=.30 | M2   | 0.100490      | 0.000490 | 99.6%     | 93.8%    |

**IMPORTANT** Singularity remained common (roughly 83%-90%). The design deliberately retained evaluable singular fits. Architecture selection followed the locked calibration rule, not warning count.

## 4. Physics v1.0 Gate 0: observer entropy to horizon entropy

Physics v1.0 proposed an information-theoretic phenomenology in which an observer subsystem has a reduced-state von Neumann entropy  $H_{\text{obs}}$  and the model also introduces a causal-horizon capacity scale. The parent paper itself marked the observer-to-horizon identification as a structural extension rather than

a microscopic derivation. Gate 0 tested whether established literature closes that bridge without adding assumptions.

## 4.1 Audit logic

6. Srednicki-type geometric vacuum entanglement supplies area scaling only after a spatial bipartition and a regulated local quantum field are specified.
7. Bombelli-Koul-Lee-Sorkin-type constructions likewise rely on geometric division and Gaussian field structure.
8. Susskind-Uglum and effective-action renormalization relate divergent matter entropy and gravitational couplings in controlled semiclassical settings, with higher-curvature dependence.
9. Matter entanglement entropy, gravitational entropy, and generalized entropy remain distinct unless a theory explicitly connects them.
10. The original  $H_{\text{obs}}$  definition supplies an observer/environment factorization but not the geometric QFT and gravitational-effective-action structures required by those bridges.

**FROZEN VERDICT** UNRESOLVED IN GENERALITY. This is not an impossibility proof. It is a stop condition: Gate 1 is not entered from the original assumptions.

The audit therefore preserved rather than upgraded the parent claim. No modified Friedmann equation, effective information stress-energy tensor, cosmological perturbation equation, or gravitational-wave spectrum was derived from Gate 0.

## 5. Appendix A: exact Hopf-bifurcation theorem

The dynamical sector is a three-variable phenomenological ODE. Its later Appendix A lineage moved from equilibrium analysis, to numerical Hopf-sign evidence, to an exact positivity theorem for the first Lyapunov coefficient over the covered positive parameter family.

### 5.1 Parent dynamical system

$$\dot{\gamma} = A v (1 - \gamma/G) - b \gamma$$

$$\dot{H} = c \gamma (M - H) - r$$

$$\dot{v} = e H_{\dot{}} + F - g v$$

For positive parent parameters and an interior physical equilibrium:

$$v^* = F/g, \quad \gamma^* = A v^* G / (bG + A v^*), \quad H^* = M - r/(c \gamma^*)$$

Define strictly positive reduced parameters  $\alpha = b + A v^*/G$ ,  $\beta = A(1-\gamma^*/G)$ ,  $\mu = c \gamma^*$ ,  $\delta = r/\gamma^*$ ,  $K = A/G$ ,  $C = c$ , with  $r > 0$  for the e-driven Hopf family.

### 5.2 Hopf surface

$$J = [[-\alpha, 0, \beta], [\delta, -\mu, 0], [e \delta, -e \mu, -g]]$$

$$\lambda^3 + S \lambda^2 + (\alpha \mu + \alpha g + \mu g - \beta \delta e) \lambda + \alpha \mu g = 0$$

$$S = \alpha + \mu + g$$

$$e_c = (\alpha+g)(\alpha+\mu)(g+\mu) / [\beta \delta (\alpha+\mu+g)]$$

$$\omega_c^2 = \alpha \mu g / (\alpha+\mu+g)$$

### 5.3 The positivity proof

The normal-form calculation uses the convention  $l_1 = \text{Re}(G_{21})/(2\omega)$  with normalized left/right eigenvectors. After imposing the Hopf identity for  $\omega^2$  and simplifying exactly, the symbolic derivation reduces the real normal-form coefficient to

$$\text{Re}(G_{21}) = N(\alpha, \mu, g, \beta, \delta, K, C) / D(\alpha, \mu, g, \beta, \delta, K, C).$$

The exported exact certificate contains 93 numerator monomials and 28 denominator monomials. Every rational coefficient in both polynomials is strictly positive. Since all seven variables are strictly positive in the theorem domain,  $N > 0$  and  $D > 0$ . Because  $\omega_c > 0$ :

$$\text{Re}(G_{21}) > 0 \Rightarrow l_1 > 0.$$

The eigenvalue crossing direction is also positive. One exact form is

$$d \text{Re}(\lambda)/de = \beta \delta S / [2(S^2 + \omega_c^2)] > 0.$$

**THEOREM** For  $\alpha, \mu, g, \beta, \delta, K, C > 0$  on the exact e-driven Hopf surface,  $l_1 > 0$ . Together with positive transversality as  $e$  increases, every covered interior e-driven Hopf bifurcation is subcritical in the stated convention.

### 5.4 Verification and scope

- Before the analytic proof, a deterministic bounded sweep produced 122,378 valid Hopf points, all with  $l_1 > 0$ .
- The v1.4.1 verifier checks positivity of all 121 exact coefficients.
- An independent 50-digit eigenvector calculation at 25 deterministic random positive points agreed with the certificate ratio  $N/D$  to a worst relative difference of  $1.53e-47$ .
- Certificate SHA-256: 34439e86255d7d9e61c483e3c97044e247d9ae09bc2c00b13791704c8a68667.
- The theorem excludes  $r=0$  ( $\delta=0$ ), where  $e$  cannot generate the same crossing.

**CLAIM FIREWALL** This is a theorem about the specified phenomenological ODE family. It does not prove that the ODE is a microscopic law, does not resolve Gate 0, and does not provide a global basin-boundary theorem.

## 6. LRSC odd-ring spectral rank theorem

The LRSC theorem concerns one explicitly defined synchronous update on an odd cyclic lattice. The strongest result is analytic: under a reflection-symmetric binary mask with at least one inactive reflection orbit, the post-stencil channel matrix has rank exactly equal to the number  $p$  of active reflection orbits.

### 6.1 Definitions

Let  $M \geq 9$  be odd,  $k = (M-1)/2$ ,  $\alpha = \pi/M$ , and use centered sites  $t = -k, \dots, k$ . Define

$$X_t = 1 + A \cos(2\pi k t / M), \quad A > 0$$

$$j_t = X_t - \min(X_{t-1}, X_t, X_{t+1}).$$

Let  $m_i \in \{0,1\}$  satisfy  $m_i = m_{-i}$ ; assume at least one reflection orbit is inactive; let  $p$  be the number of active reflection orbits. The real conjugate-balanced channel columns are mask-supported cosine modes

scaled by the Fourier coefficients of  $j$ . A circulant stencil  $S$  has Fourier multiplier  $s_q = \cos(2\pi q/M) - 1$ , and  $V = SW$ .

**THEOREM** Under these assumptions,  $\text{rank}(V) = p$ .

## 6.2 Proof map

11. Exact shedding formula. The near-Nyquist carrier can be written  $X_t = 1 + A(-1)^t \cos(\alpha t)$ . This gives  $j_t = 0$  on odd sites,  $j_0 = A(1 + \cos \alpha)$ , and an explicit cosine formula on nonzero even sites.
12. Non-vanishing low Fourier coefficients. The finite cosine progression for the Fourier coefficients of  $j$  is summed exactly. Separate residue-class formulas for  $M = 4L + 3$  and  $M = 4L + 1$  show the relevant brackets cannot vanish for  $0 \leq r < k$ . Hence the first  $k$  Fourier scales required by the rank argument are nonzero.
13. Pre-stencil rank. Choose one representative  $i_s$  from each active reflection orbit and set  $x_s = \cos(2\pi i_s/M)$ . The  $p \times p$  active-row / first- $p$ -channel submatrix is, up to nonzero scales,  $[T_r(x_s)]$ , where  $T_r$  is the Chebyshev polynomial of degree  $r$ . Its determinant is a nonzero leading-coefficient factor times the Vandermonde product  $\prod_{u < v} (x_v - x_u)$ , because the  $x_s$  are distinct. Thus  $\text{rank}(W) \geq p$ . Reflection symmetry and mask support give  $\text{rank}(W) \leq p$ , so  $\text{rank}(W) = p$ .
14. Stencil preserves rank. The multiplier  $s_q$  vanishes only at  $q = 0$ , so  $\ker(S) = \text{span}\{1\}$ . Every vector in  $\text{span}(W)$  vanishes on every inactive site, and therefore no nonzero constant lies in  $\text{span}(W)$ . Hence  $\ker(S)$  intersect  $\text{span}(W) = \{0\}$ ;  $S$  is injective on the channel span;  $\text{rank}(V) = \text{rank}(SW) = p$ .

## 6.3 Independent checks

| Check                                  | Recorded outcome                                                 |
|----------------------------------------|------------------------------------------------------------------|
| Odd rings                              | $M = 9, 11, \dots, 401$ : 197 rings                              |
| Fourier coefficients                   | 20,094 coefficients checked                                      |
| Largest direct-sum vs compact-form gap | $< 2e-14$                                                        |
| Largest direct-DFT vs compact-form gap | $< 2e-14$                                                        |
| Rank cases                             | 35/35 pass for all $p$ in selected $M$ values                    |
| Locked $M = 99$ benchmark              | 41 active reflection orbits; theorem gives $\text{rank}(V) = 41$ |

**CLAIM FIREWALL** The theorem is about a single-step finite cyclic operator. It does not prove a microscopic physical law, universal compression dimension, multi-step dynamics, or Navier-Stokes regularity.

## 7. LRSC fixed-benchmark compression certificate

A separate reproducibility supplement asked a finite computational question: for one locked  $M = 99$ ,  $k_{\text{idx}} = 49$ ,  $\theta = 0.08$ ,  $A = 0.50$  benchmark, how many of the 50 Fourier channels are needed to reconstruct the one-step update to relative complex-L2 error at most 0.001?

The operator closure using all 50 channels was  $2.3808015585084204e-16$ . Exact-cardinality subsets  $K = 1$  through  $K = 10$  were exhaustively enumerated in IEEE binary64 arithmetic.  $K = 11$  required only a passing witness once every smaller cardinality had failed.

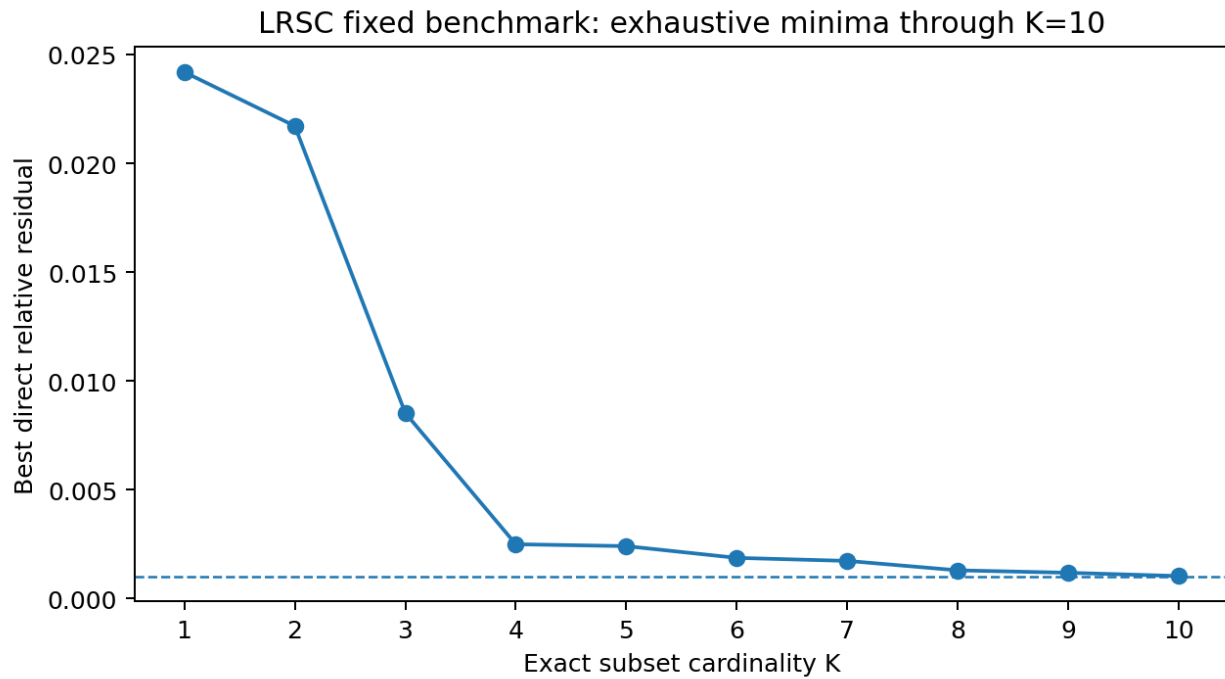


Figure 2. Best exhaustive residual through  $K=10$ . The dashed line is the 0.001 threshold.

| K  | Subsets visited    | Best subset                     | Direct residual | Pass? |
|----|--------------------|---------------------------------|-----------------|-------|
| 1  | 50                 | {49}                            | 0.0241917969    | No    |
| 2  | 1,225              | {48,49}                         | 0.0217153914    | No    |
| 3  | 19,600             | {0,1,49}                        | 0.0085289394    | No    |
| 4  | 230,300            | {0,1,48,49}                     | 0.0025003115    | No    |
| 5  | 2,118,760          | {0,1,3,48,49}                   | 0.0024131940    | No    |
| 6  | 15,890,700         | {0,1,46,47,48,49}               | 0.0018745071    | No    |
| 7  | 99,884,400         | {0,1,45,46,47,48,49}            | 0.0017353561    | No    |
| 8  | 536,878,650        | {0,1,44,45,46,47,48,49}         | 0.0013025959    | No    |
| 9  | 2,505,433,700      | {0,1,43,44,45,46,47,48,49}      | 0.0011898947    | No    |
| 10 | 10,272,278,170     | {0,1,42,43,44,45,46,47,48,49}   | 0.0010398206    | No    |
| 11 | witness sufficient | {0,1,3,42,43,44,45,46,47,48,49} | 0.0009623691    | Yes   |

**NUMERICAL CERTIFICATE** For the stated benchmark and binary64 implementation: no subset of size  $\leq 10$  passes, and a size-11 witness passes. Therefore the minimum cardinality is 11 for this numerical benchmark. This is not a rounding-error-enclosed interval proof.

Earlier proposed  $K=7$  passes were explicitly falsified. Two disputed  $K=7$  candidates reconstruct to residuals 0.0029954428 and 0.0018748270, respectively, and neither is the exhaustive  $K=7$  minimum. The corrected reproducibility supplement preserves those failed candidates in the audit record rather than deleting them.

## 8. Navier-Stokes bridge audit: what survives and what fails

The Navier-Stokes line began by asking whether finite-ring spectral structure or the phenomenological ODE could transfer a useful mechanism to the three-dimensional incompressible periodic equation. The

first result was negative: the transfer is not automatic. The useful outcome was a cleaner separation between finite computational identities, false energy-only bounds, and one valid small-data estimate.

## 8.1 Exact finite Galerkin identities

$$\begin{aligned} dE/dt &= -\nu \sum_k |k|^2 |a_k|^2 \\ \Omega &= (1/2) \sum_k |k|^2 |a_k|^2, \quad D = \sum_k |k|^4 |a_k|^2 \\ d\Omega/dt &= T - \nu D \end{aligned}$$

The code explicitly enforces incompressibility and conjugate symmetry, applies the Leray projector, and sums all ordered convolution pairs. A physical-space cubic-gradient calculation agreed with the Fourier enstrophy transfer to about  $4.4e-15$  in the seeded check. This corroborates finite algebra; it is not an infinite-dimensional proof.

## 8.2 Two overstrong flux bounds are false

15. Amplitude scaling: for a fixed sparse triad family, nonlinear flux scales like  $A^3$  while energy scales like  $A^2$ . Therefore a universal inequality  $|P_{i,K}| \leq C E$  with  $C$  independent of amplitude cannot hold.
16. Frequency scaling:  $u(x) \rightarrow u(mx)$  leaves energy invariant under the rescaling family while the relevant flux grows like  $m$ . Therefore  $|P_{i,K}| \leq C E^{(3/2)}$  cannot hold with  $C$  independent of cutoff/frequency.

These counterexamples do not reject derivative-dependent, viscosity-dependent, or data-dependent estimates. They only close two tempting but overly strong routes.

## 8.3 Surviving small-data theorem

Let  $G_N = \|\text{grad } u_N\|_2^2$  and  $D_N = \|\Delta u_N\|_2^2$ . After subtracting the mean velocity, Fourier Cauchy-Schwarz gives

$$\|u_N - \text{mean}(u_N)\|_\infty \leq (\sum_{k \neq 0} |k|^{-4})^{(1/2)} \|\Delta u_N\|_2.$$

Counting lattice points by maximum coordinate gives at most  $24n^2+2$  points on the  $n$ -th cube shell, so

$$\sum_{k \neq 0} |k|^{-4} \leq \sum_{n \geq 1} (24n^2+2)/n^4 = 4\pi^2 + \pi^4/45 = C_0^2.$$

$$C_0 = \sqrt{4\pi^2 + \pi^4/45} \text{ approximately } 6.45314.$$

Hölder then yields the cutoff-independent nonlinear estimate

$$|T_N| \leq C_0 \sqrt{G_N} D_N,$$

and hence

$$d(G_N/2)/dt \leq -(\nu - C_0 \sqrt{G_N}) D_N.$$

If  $C_0 \|\text{grad } u_0\|_2 < \nu$ , continuity prevents  $G_N$  from crossing upward through its initial level. The usual Galerkin/strong-solution continuation then yields global smoothness for this small-data class.

**PROVED SUBCASE** This is a sufficient small-data result, classical in spirit and intentionally conservative. It does not settle arbitrary smooth initial data.

## 9. WP16 K36 finite mechanism and the N14 prospective gate

Later work isolated a finite output-channel mechanism around the triad  $p=(3,2,2)$ ,  $q=(3,-2,1)$ ,  $k=(6,0,3)$ . The key methodological feature is that the 36-source coalition K36 was frozen from N11 before the N12 and N13 holdouts were generated.

### 9.1 Prospective static holdouts

For each state, the frozen coalition had to satisfy all three conditions: same sign as the full k-channel, at least 90% grouped absolute source mass, and signed share in  $[0.80,1.20]$ . All three N12 states and all three N13 states passed.

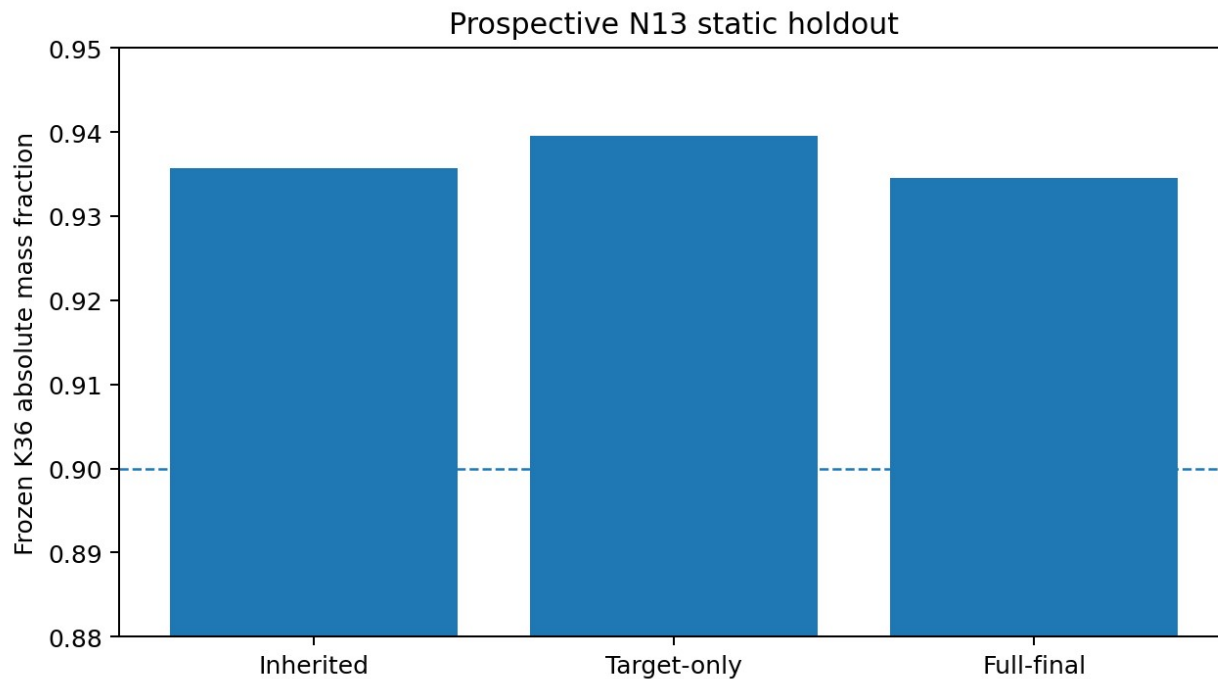


Figure 3. N13 frozen-K36 absolute mass fractions; all three exceed the predeclared 0.90 gate.

| N13 state   | Mass fraction | Signed share | Same sign | Static gate |
|-------------|---------------|--------------|-----------|-------------|
| Inherited   | 0.935731      | 1.029876     | Yes       | PASS        |
| Target-only | 0.939559      | 1.024681     | Yes       | PASS        |
| Full-final  | 0.934538      | 1.025928     | Yes       | PASS        |

### 9.2 What the post-hoc stress tests falsified

- A fixed K36 90% theorem for all divergence-free Galerkin states is false: an explicit N7 counterexample can make the frozen coalition arbitrarily weak by amplifying an outside source pair while preserving the tracked anchor modes.
- Phase-only robustness does not imply trajectory persistence. At fixed magnitudes/polarizations there is a deterministic local phase-defect neighborhood, but actual evolution changes magnitudes and vector polarizations too.
- Under the finite Galerkin ODE with  $\nu=0.1$ , all six N12/N13 states still pass through  $t=0.001$ , but the 90% mass criterion later fails at sampled times about 0.0020-0.0026.

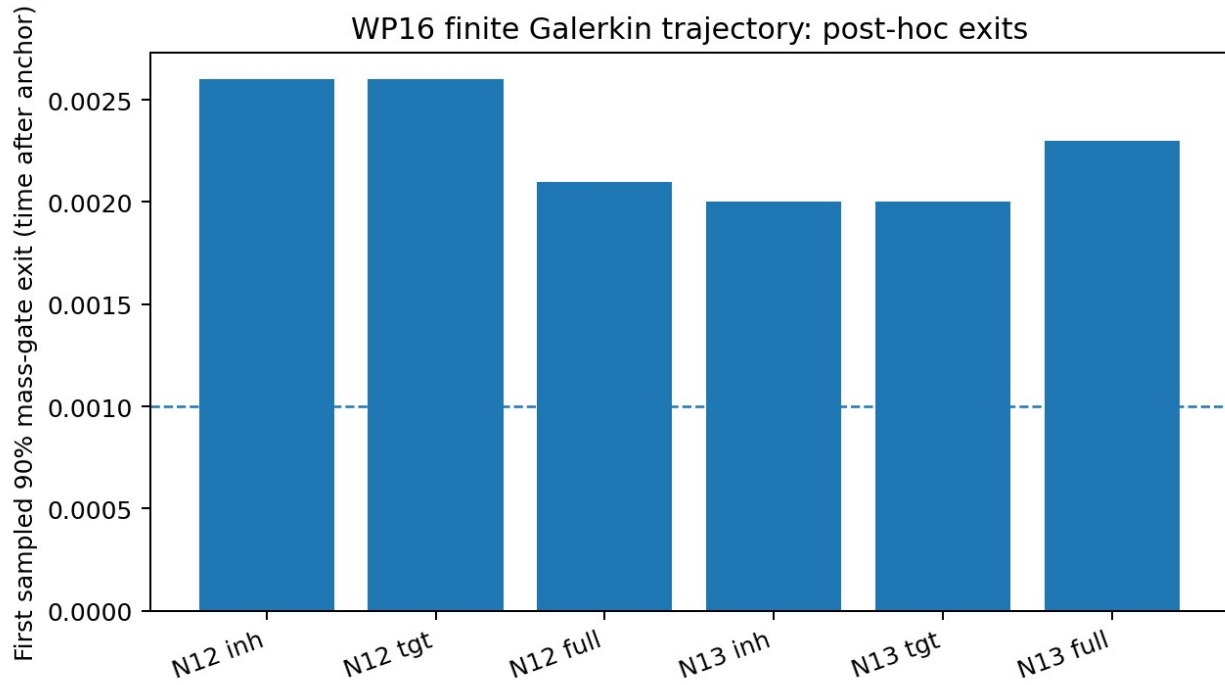


Figure 4. First sampled post-hoc 90% mass-gate exit times for the six N12/N13 trajectories.

### 9.3 Margin dynamics

Let  $I$  be the grouped absolute contribution inside K36 and  $O$  the grouped absolute contribution outside it. The 90% mass condition is equivalent to

$$F = I - 9 O \geq 0.$$

At the six optimized anchors, directional diagnostics found  $dF/dt$  positive (+121,277 to +401,330), with scalar phase helping the margin. At the first sampled exits, the full rates were negative (-239,458 to -522,435), and both radial-magnitude and vector-polarization components were adverse in all six. Viscosity contributed a small positive margin rate. The decomposition residual was below  $2.85e-4$  in absolute rate units.

### 9.4 Exact finite rate inequality

For each ordered source  $l+r=k$ , define the projected source  $d_{lr}$  and normalized contribution  $x_{lr}=C_{lr}/z$ , where  $z$  is the tracked nonzero complex channel coefficient. Differentiating the numerator and denominator along the finite Galerkin velocity  $v=\dot{a}$  gives

$$\dot{x}_{lr} = \dot{C}_{lr} / z - C_{lr} \dot{z} / z^2.$$

If  $c_g$  is the sum of  $\text{Im}(x_{lr})$  over a source-orbit group  $g$ , then  $|d|c_g|/dt|$  is bounded by the sum of  $|\dot{x}_{lr}|$  in that group, with the same lower one-sided control at zero groups by local Lipschitzness. Writing  $E_I$  and  $E_O$  for the corresponding source-derivative envelopes gives the integrated finite inequality

$$F(t) \geq F(0) - \int_0^t [E_I(s) + 9 E_O(s)] ds, \quad \text{while } z(s) \neq 0.$$

A broader norm estimate is available using convolution Cauchy-Schwarz:

$$S_0 \leq \|a\|_2 \|\text{grad } a\|_2$$

$$S_1 \leq \|v\|_2 \|\text{grad } a\|_2 + \|a\|_2 \|\text{grad } v\|_2$$

$$E_I + 9E_O \leq (9w/|z|) \{ |B|S_1 + [|\dot{B}| + |B||\dot{z}/z|]S_0 \}.$$

This estimate is cutoff-independent only conditional on the displayed norms, the nonvanishing denominator, and integrability. On the archived N12/N13 anchors it is far too loose to explain persistence: the broad H1 bound is about 185-208 times the already conservative exact source envelope. That failure is itself informative: cancellation and polarization geometry carry essential information lost by the norm bound.

## 9.5 N14: prospective before outcome

Before any N14 result was generated, the next holdout was frozen with immutable K36 keys, the same static criteria,  $\nu=0.1$ , an alias-safe selection grid 48, seed 20260939, and a 520-proposal search schedule. The memory-bounded implementation preserves the original objective and uses atomic checkpoints. The time gate is fixed at  $dt=0.0001$  through  $t=0.0030$  with a half-step crossing check and the same margin-velocity decomposition.

**STATUS OF THIS PDF** No N14 scientific outcome is included here. At the time this synthesis was assembled, the prospective N14 continuation was being executed. The eventual result must be appended as a separate dated update, not back-filled into this edition.

# 10. Public reproducibility ledger and references

The research record is intentionally versioned. The public reader should treat the following primary artifacts as the controlling sources for the claims summarized above.

## 10.1 Primary author releases

- **Self-Referential Processing Pre-Registration v8.6.3:** <https://doi.org/10.17605/OSF.IO/37H2Y>
- **Physics v1.0 parent project / Gate 0 lineage:** <https://doi.org/10.17605/OSF.IO/TSW4B>
- **Appendix A v1.4 analytic positivity theorem:** <https://doi.org/10.17605/OSF.IO/U7M6H>
- **LRSC Odd-Ring Spectral Rank Theorem v1.2:** <https://doi.org/10.17605/OSF.IO/NM5BW>
- **Navier-Stokes Bridge Audit repository:** <https://github.com/reggaesharkk/navier-stokes-bridge-audit>
- **Research navigation / universe mirror:** <https://github.com/reggaesharkk/Reggae-shark-universe->

## 10.2 Selected frozen hashes

| Artifact                                | SHA-256 / identifier                                             |
|-----------------------------------------|------------------------------------------------------------------|
| Self-referential item pool              | 128ff78bb6beb1640fa51897ab97dbb9d07dc56de276973d05a48dda524b53d3 |
| Appendix A exact positivity certificate | 34439e86255d7d9e61c483e3c97044e247d9eae09bc2c00b13791704c8a68667 |
| WP16 N13 phase continuation             | 13e5e56676b9398e7c7ec32f55e32a4d07dec5a3830c67787c6cd6fb5c8e59cd |
| DesignLab Phase 7 results.csv           | cce97faba969a6da873d9a5a34d9da6421fdf24d1a04df29c1a599fb3044e3d0 |

### 10.3 Scientific claim ledger

| Claim                                                                                      | Status                             | Why                                                                                                  |
|--------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------|
| Behavioral self-referential effect exists in live AI systems                               | OPEN                               | The protocol is designed to test it; recovery/simulation evidence is not live confirmatory evidence. |
| M1 GLMM architecture is calibrated under the locked Phase 7 DGP                            | SIMULATION VALIDATED               | Prespecified exact-binomial screens passed at $N=500$ for both tested $\rho$ values.                 |
| Generic observer entropy equals horizon/gravitational entropy                              | UNRESOLVED                         | Gate 0 found the necessary geometric/QFT/gravitational bridge absent from the parent assumptions.    |
| Covered e-driven Hopf family has $l_1 > 0$                                                 | PROVED WITHIN MODEL                | Exact symbolic reduction to positive-coefficient $N/D$ plus positive transversality.                 |
| LRSC odd-ring post-stencil rank equals active reflection-orbit count $p$                   | PROVED                             | Nonvanishing low modes + Chebyshev/Vandermonde rank + stencil injectivity.                           |
| LRSC fixed $M=99$ benchmark has $K_{0.001}=11$                                             | NUMERICAL CERTIFICATE              | Exhaustive binary64 search for $K \leq 10$ and direct $K=11$ passing witness.                        |
| Simple energy-only bounds control all Navier-Stokes low-shell transfer                     | FALSIFIED                          | Amplitude and frequency scaling counterexamples.                                                     |
| Small periodic data with $C_0    \text{grad } u_0   ^2 < \nu$ stay globally $H^1$ -bounded | PROVED SUBCASE                     | Cutoff-independent lattice-sum estimate and continuation argument.                                   |
| Frozen K36 gives $\geq 90\%$ source mass for all Galerkin states                           | FALSIFIED                          | Explicit N7 outside-source counterexample.                                                           |
| Frozen K36 transfers statically from N11 to N12/N13 optimized states                       | PROSPECTIVE FINITE EVIDENCE        | All six preregistered static holdouts passed.                                                        |
| Frozen K36 remains $\geq 90\%$ along those finite trajectories indefinitely                | FALSIFIED FOR SAMPLED TRAJECTORIES | All six sampled trajectories later exit the mass gate.                                               |
| Arbitrary-data 3D unforced Navier-Stokes global regularity is proved here                  | NO                                 | No such claim is made; the main arbitrary-data problem remains outside these results.                |

### 10.4 Selected external references used in the source program

- Azaria, A. & Mitchell, T. (2023). The internal state of an LLM knows when it is lying. arXiv:2304.13734.
- Marks, S. & Tegmark, M. (2024). The Geometry of Truth. COLM / arXiv:2310.06824.
- Meng, K. et al. (2022). Locating and editing factual associations in GPT. NeurIPS 35, 17359-17372.
- Olsson, C. et al. (2022). In-context learning and induction heads. arXiv:2209.11895.
- von Oswald, J. et al. (2023). Transformers learn in-context by gradient descent. ICML, PMLR 202:35151-35174.
- Srednicki, M. (1993). Entropy and area. Physical Review Letters 71, 666-669.
- Bombelli, L., Koul, R. K., Lee, J., & Sorkin, R. D. (1986). A quantum source of entropy for black holes. Physical Review D 34, 373-383.
- Clay Mathematics Institute / Fefferman problem statement for Navier-Stokes existence and smoothness.

## Closing statement

The research program is best understood not as one giant theory but as a sequence of increasingly disciplined tests. Several attractive interpretations were rejected. Some finite mechanisms survived only under narrower conditions. Two exact analytic theorems emerged inside explicitly defined mathematical models. One conservative Navier-Stokes small-data argument survived the bridge audit. The AI work remains a preregistered empirical question. The strongest common result across the projects is methodological: claim boundaries became sharper as the work progressed.

**PUBLIC READING NOTE** Readers are encouraged to reproduce, challenge, and improve the work. When citing a theorem or numerical result, cite the corresponding frozen primary release rather than this synthesis alone.